

A Deep Learning Approach for Dynamic Modeling of Stimulated Raman Scattering in Chalcogenide Microstructured Optical Fibers

Jun Fu¹, Senior Member, IEEE, Yupeng Liu, Qi Wang², Meiting Pan³, and Tonglei Cheng⁴

Abstract—Stimulated Raman scattering (SRS) plays a pivotal role in applications such as optical communications, fiber optic sensing, and spectral analysis. However, traditional modeling methods like the split-step Fourier method (SSFM) are computationally demanding. In response to these challenges, we propose a novel deep learning framework based on a hybrid neural network, specifically architected to capture the complex spatio-temporal dependencies inherent in nonlinear pulse propagation. Our model offers rapid and precise predictions of SRS behavior, alleviating the need for computationally expensive simulations like SSFM. To validate the model's performance, we conducted experiments using chalcogenide microstructured optical fibers (MOFs), which are attracting attention due to their high Raman gain coefficient and wide spectral range in the mid-infrared (MIR) region. Specifically, we demonstrate the first successful generation of MIR SRS in a 2- μm direct-pumped suspended-core As_2S_3 MOF, which provides a crucial real-world dataset for model validation. The results demonstrate that our hybrid neural network is 116 times faster on a GPU and 44 times faster on a CPU compared to SSFM while maintaining accuracy and generalization. This significant acceleration paves the way for real-time analysis and inverse design of nonlinear photonic systems, tasks previously intractable with traditional methods.

Index Terms—Hybrid neural network, microstructured optical fibers (MOFs), split-step Fourier method (SSFM), stimulated Raman scattering (SRS).

I. INTRODUCTION

RESEARCH and applications in the realm of optical fibers are continuously expanding, with stimulated Raman scattering (SRS) emerging as a pivotal nonlinear optical effect garnering widespread attention [1], [2]. SRS fundamentally describes the interaction between light and the vibrational modes of molecules within a medium [3]. During this inelastic scattering process, an incident pump photon excites a molecule to a virtual energy state, which then de-excites by emitting

a lower frequency Stokes photon and creating a molecular vibration (an optical phonon). When the pump light is sufficiently intense, this process becomes stimulated; the presence of Stokes photons triggers a cascading amplification, leading to exponential growth of the Stokes wave and significant power transfer from the pump [4]. This power transfer evolves dynamically along the fiber. Above a certain threshold power, the Stokes wave grows exponentially, depleting the pump [5]. Further propagation can lead to cascading effects, generating higher order Stokes waves, and potentially anti-Stokes waves through processes like four-wave mixing [6]. The resulting spectral evolution is complex, depending critically on pump power, pulse duration, fiber length, and the fiber's specific dispersive and nonlinear properties. Accurately modeling this intricate dynamic process is essential for many applications, yet remains a challenge.

While SRS in conventional silica fibers is well-studied in the visible and near infrared, research in the mid-infrared (MIR) domain has been hindered by the high losses of traditional fiber materials. To overcome this barrier, chalcogenide optical fibers have emerged as a leading solution. Compared to silica, chalcogenide glasses offer a broader transmission window and a significantly higher nonlinear material index, making them ideal for MIR applications. Notably, materials such as As_2S_3 and As_2Se_3 are widely used due to their wide transmission up to 12 or 15 μm and high nonlinearity [7], [8], [9], [10], [11]. However, the zero-dispersion wavelengths (ZDWs) of bulk As_2S_3 and As_2Se_3 are located far into the MIR (4.8 and 8 μm , respectively) [12], which complicates efficient nonlinear interactions with common pump sources. This challenge is directly addressed by chalcogenide microstructured optical fibers (MOFs) [13], [14], [15]. MOFs utilize an air-hole-structured cladding to provide strong waveguide confinement. This structure not only yields a high effective nonlinearity but also allows the fiber's dispersion profile, including the ZDW, to be precisely engineered [16], [17], [18].

The propagation dynamics of SRS in these fibers are governed by the coupled nonlinear Schrödinger equation (CNLSE) [19], which is a nonlinear partial differential equation that does not generally lend itself to analytic solutions. Therefore, to verify the accuracy of experimental data and better optimize experimental parameters, numerical methods are usually required to understand the nonlinear effects in optical fibers. Traditional numerical simulation methods often rely on the

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Jun Fu is with the State Key Laboratory of Synthetical Automation for Process Industries, Northeastern University, Shenyang 110819, China (e-mail: junfu@mail.neu.edu.cn).

Yupeng Liu is with the State Key Laboratory of Synthetical Automation for Process Industries and the School of Future Technologies, Northeastern University, Shenyang 110819, China (e-mail: 2210336@stu.neu.edu.cn).

Qi Wang, Meiting Pan, and Tonglei Cheng are with the College of Information Science and Engineering, Northeastern University, Shenyang 110819, China (e-mail: 2170930@stu.neu.edu.cn; 2200772@stu.neu.edu.cn; chengtonglei@ise.neu.edu.cn).

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split-step Fourier method (SSFM) to iteratively solve the CNLSE [4]. However, iteratively solving the CNLSE with the SSFM requires a small step size for accuracy, making this approach computationally demanding and creating significant bottlenecks when designing or optimizing experiments in real time. To address this computational challenge, deep learning has recently emerged as a powerful alternative, offering the potential to create fast and accurate surrogate models [20], [21]. These approaches generally fall into two categories: data-driven methods that learn from simulation ensembles [20], [22], [23] and physics-informed neural networks (PINNs) [24], [25], [26] that embed the governing equations into the learning process. A detailed comparison of our work with the state of the art in both areas is presented in Section II. The main contributions of this work are summarized as follows.

- 1) We propose and validate a novel hybrid deep learning architecture that synergistically combines a convolutional neural network (CNN), a bidirectional long short-term memory (BiLSTM) network, and an attention mechanism. This architecture is specifically designed to effectively capture the complex spatio-temporal features inherent in the SRS process.
- 2) We apply this model to a new and challenging domain: the dynamic modeling of MIR SRS in chalcogenide suspended-core MOFs. Crucially, we provide the first experimental demonstration of MIR SRS in an As_2S_3 MOF via direct $2\ \mu\text{m}$ pumping and use this real-world data to validate our model's predictive accuracy.
- 3) We demonstrate that our proposed framework serves as a highly efficient surrogate for traditional numerical solvers, achieving a computational speedup of $116\times$ on a GPU and $44\times$ on a CPU. This model maintains high accuracy and exhibits strong generalization, evidenced by its ability to predict SRS evolution under varying pump powers and fiber lengths for As_2S_3 . We further confirm the robustness and applicability of our architecture by successfully applying it to model SRS in a physically distinct As_2Se_3 MOF.

II. RELATED WORKS

Recent years have seen a surge in applying deep learning to model nonlinear fiber dynamics, with research primarily following two distinct paths: data-driven models that learn from large simulation datasets, and PINNs, which incorporate the governing equations directly into the network's loss function.

Data-driven models, particularly those using recurrent neural networks (RNNs), have proven effective. The seminal work by Salmela et al. [20] demonstrated that an LSTM network could predict complex dynamics like supercontinuum generation orders of magnitude faster than traditional solvers. This approach was further developed by others, such as Pu et al. [22] and Fang et al. [23], who employed LSTM and BiLSTM architectures to successfully model soliton dynamics in mode-locked lasers. While these works established the viability of RNNs, they often relied on a single type of recurrent architecture. In contrast, our work proposes a more sophisticated hybrid architecture. We combine a CNN as a powerful feature extractor for the high-dimensional spectral

data, which is then fed into a BiLSTM to capture the evolutionary dynamics along the fiber. An attention mechanism further refines the model's focus on critical spectral regions during the evolution, a synergistic combination designed specifically for the complex features of SRS.

In parallel, PINNs have emerged as a powerful alternative that reduces the dependency on large datasets [24]. By embedding the NLSE into the loss function, PINNs ensure physical consistency. Recent advancements, such as the SRS-Net for Raman amplifier modeling by Song et al. [25] and the enhanced PINN for high-frequency dynamics by Jiang et al. [26], highlight the progress in this area. However, PINNs can struggle with the "curse of dimensionality" and may face challenges in learning dynamics with high non-linearity or when the governing equations do not perfectly capture all experimental nuances. Our data-driven approach, while requiring an initial data generation phase, may be more robust for modeling novel systems like our As_2S_3 MOF, where subtle, higher order effects present in experiments can be learned directly from the data.

Finally, model efficiency and advanced architectures are also active research areas. Sui et al. [27] proposed a lightweight CNN to balance speed and accuracy for multipulse propagation. More advanced strategies, such as the ensemble learning and knowledge distillation framework proposed by Xu et al. [21], aim to compress large, accurate "teacher" models into efficient "student" models. Our work carves out a distinct niche. While not as lightweight as a pure CNN, our hybrid model is specifically tailored to the complexity of SRS, achieving a significant, experimentally validated speedup without oversimplifying the architecture. It provides a direct, end-to-end solution without the multistage complexity of knowledge distillation.

III. METHODS

Our proposed deep learning framework for modeling SRS dynamics is illustrated in Fig. 1. The core of this framework is a hybrid neural network that learns to predict the complex spectral evolution from a set of initial conditions. The overall workflow consists of two primary stages: data generation and model prediction.

First, to create a comprehensive dataset for training, we use the SSFM to numerically solve the CNLSE, simulating the SRS process under a wide range of conditions. Second, we collect real-world experimental data from a custom-built As_2S_3 MOF platform, which serves as an essential validation set to verify the model's performance in a practical scenario. The hybrid neural network is then trained on the simulation data to act as a surrogate for the computationally expensive SSFM. Sections III-A–III-D provide a detailed description of each component of this hybrid architecture.

A. Convolutional Neural Network

CNNs, introduced in 1989 [28], are specialized neural networks highly effective at processing grid-like data, such as images. Their architecture is foundational to modern computer vision and has been successfully adapted for sequence analysis

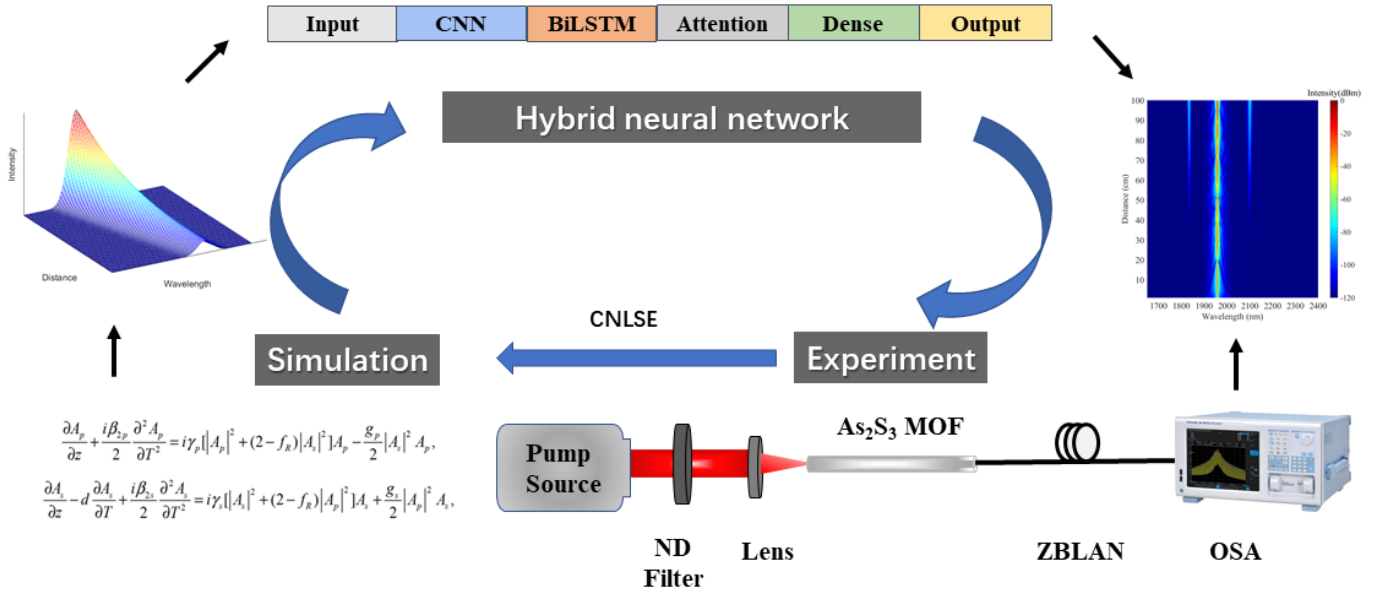


Fig. 1. Flowchart of the deep learning framework for modeling the MIR SRS effect in chalcogenide MOFs. The proposed hybrid neural network utilizes simulation results for training and testing, while data from the experimental platform serve as the validation dataset. The simulation process employs the SSFM to iteratively solve the CNLSE.

[29]. The core components of a CNN are the convolutional layer and the pooling layer. The convolutional layer utilizes filters (or kernels) that slide across the input data, performing convolution operations to create feature maps. This operation leverages two key principles: local connectivity (each neuron responds only to a local region of the input) and weight sharing (the same filter is applied across the entire input). This design significantly reduces the number of trainable parameters compared to fully connected networks, making the model more efficient and mitigating overfitting.

For a 2-D input I (such as an image) and a kernel K , the convolution operation $(I * K)$ at position (i, j) is defined as

$$(I * K)(i, j) = \sum_m \sum_n I(i - m, j - n) K(m, n). \quad (1)$$

This operation extracts local patterns. Following the convolution, a pooling layer (e.g., average or max pooling) downsamples the feature maps, reducing their dimensionality while retaining the most salient information.

B. Bidirectional Long Short-Term Memory

RNNs are primarily designed for processing sequential data, retaining relevant information from past inputs to predict future outputs. However, they struggle with capturing long-term dependencies effectively. To address this limitation, LSTMs were introduced [30], which employ a sophisticated memory cell (Fig. 2) governed by three gates (input, forget, and output gates) that selectively manage the flow of information, allowing the network to retain or discard data over long sequences [31], [32], [33], [34].

A conventional LSTM processes information solely in a forward direction, drawing insights from past data, as in Fig. 2. In contrast, BiLSTMs incorporate two LSTM layers: one processing input in the forward direction and the other

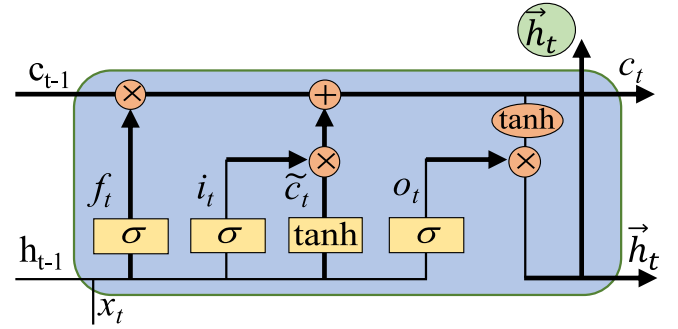


Fig. 2. Schematic of a single LSTM cell.

in the backward direction. This design allows the forward LSTM to capture past data information, while the backward LSTM gathers future data insights [35]. The outputs of these two hidden layers are then merged to enhance prediction accuracy. The workflow of a BiLSTM involves processing the input sequence in both forward and backward directions simultaneously, leveraging past and future data to improve the model's understanding and predictive capabilities. The operation of a BiLSTM is as follows:

$$\begin{aligned} i_t &= \sigma(W_i * x_t + R_i * h_{t-1} + b_i) \\ o_t &= \sigma(W_o * x_t + R_o * h_{t-1} + b_o) \\ f_t &= \sigma(W_f * x_t + R_f * h_{t-1} + b_f) \\ \tilde{c}_t &= \tanh(W_c * x_t + R_c * h_{t-1} + b_c) \\ c_t &= f_t \odot c_{t-1} + i_t \odot \tilde{c}_t \\ \vec{h}_t &= o_t \odot \tanh(c_t) \\ h_t &= \sigma(\vec{h}_t \oplus \overleftarrow{h}_t) \end{aligned} \quad (2)$$

where σ is the sigmoid activation function, $\tanh$ is the hyperbolic tangent activation function, and $\odot$ is the element-wise

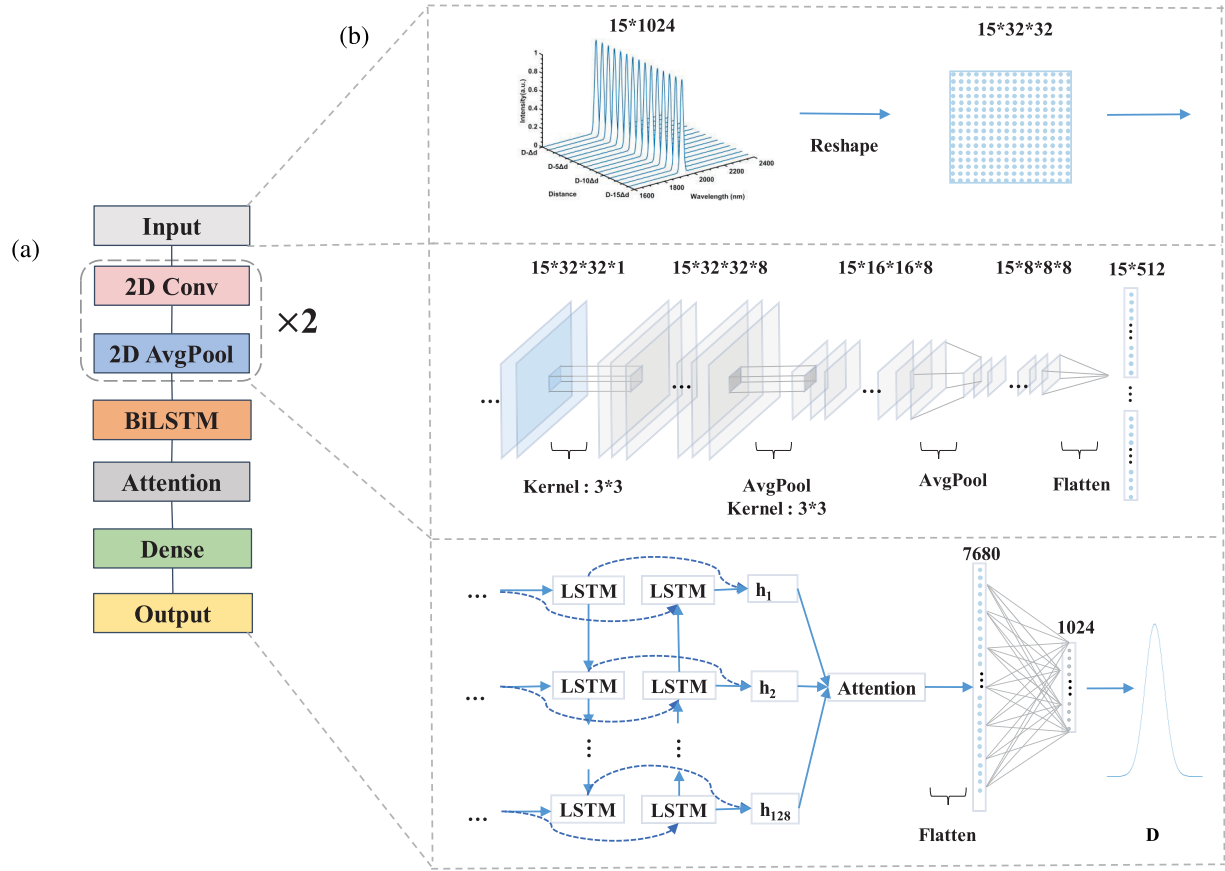


Fig. 3. Hybrid neural network. (a) Schematic of the hybrid neural network architecture used, illustrating the input layer, two convolutional layers, two average pooling layers, the BiLSTM layer, the attention layer, the hidden (dense) layer, and the output layer. (b) Detailed parameter settings and corresponding data dimensions for each layer of the network. The neural network utilizes the 15 previous spectral intensity profiles from distance $D - 15\Delta d$ to $D - \Delta d$ in the evolution to predict the subsequent spectrum at distance D . Refer to Supplementary Material for further elucidation on the principles of each layer of the network.

multiplication operation. W_i , W_o , W_f , and W_c refer to the weight matrices associated with the current input x_t . Similarly, R_i , R_o , R_f , and R_c are the weight matrices related to the previous state h_{t-1} . The terms b_i , b_o , b_f , and b_c represent the bias values. Finally, h_t represents the output of both the forward LSTM and backward LSTM units, and $\oplus$ denotes the element-wise summation.

C. Attention Mechanism

To further refine the model's focus, we apply an attention mechanism to the output of the BiLSTM layer [36], [37], [38]. In the context of sequence modeling, the attention mechanism allows the model to dynamically assign "importance" weights to different time steps in the BiLSTM's output sequence.

This selective focus is crucial for modeling SRS, as it enables the model to concentrate on the most pivotal regions of spectral change (e.g., the emergence of a new Stokes order) when making a prediction. The mechanism computes a final context vector s as a weighted sum of the BiLSTM output vectors h_t , as defined in the following equation:

$$\begin{aligned} s_t &= \tanh(W_h h_t + b_h) \\ a_t &= \frac{\exp(s_t)}{\sum_t \exp(s_t)} \\ s &= \sum_t a_t h_t \end{aligned} \quad (3)$$

where h_t represents the feature vector obtained from the BiLSTM layer at time t . W_h and b_h are the weight matrix and bias vector, respectively, that are initialized during neural network training, while s_t denotes the attention score. Then, a_t represents the normalized attention weight obtained by applying the softmax function to each attention score. Finally, s is the final context vector, which effectively captures the most relevant features from the entire sequence. This vector is then passed to the final dense layer for prediction.

D. Hybrid Neural Network Architecture

The architecture of the hybrid neural network model employed for predicting SRS is depicted in Fig. 3. This model's design is a synergistic integration of the three components described above (CNN, BiLSTM, and Attention), each chosen to address a specific aspect of the SRS dynamics.

This integration begins by treating the input data as a 2-D spatio-temporal task. Our input consists of 15 consecutive spectral profiles. Each profile is first downsampled using a 0.75-nm FWHM super-Gaussian spectral filter, resulting in 1024 equally spaced intensity bins spanning from 1640 to 2408 nm. By reshaping this 15×1024 input into a $15 \times 32 \times 32$ tensor [as illustrated in Fig. 3(b)], the architecture can effectively assign specific roles to each component.

This representation allows the hybrid architecture to function as follows.

First, a 2-D convolutional block acts as a powerful front-end feature extractor. This 2-D CNN performs a function that standard components in isolation cannot; it simultaneously learns local correlations along two critical axes as follows.

- 1) *The Spatial (Frequency) Axis*: The convolutional kernels learn to recognize patterns within each individual spectrum, such as the shape, width, and emergence of Raman peaks.
- 2) *The Temporal (Propagation) Axis*: The kernels also learn how these spectral patterns evolve over short propagation distances within the 15-step input window.

This block, consisting of two 2-D convolutional layers (8 filters, 3×3 kernel) and 2-D average pooling layers, acts as a highly efficient feature engineering step, abstracting the raw data into a condensed sequence of meaningful feature vectors.

Second, this sequence of features extracted by the CNN is processed by the BiLSTM layer. This component is essential for modeling the causal, step-by-step evolution of the spectrum along the entire fiber length. It captures the long-range temporal dependencies that the CNN, with its local receptive field, cannot. The BiLSTM layer contains 256 neurons (128 for the forward and 128 for the backward pass).

Finally, the attention mechanism is applied to the BiLSTM's output. Its purpose is to allow the model to dynamically weigh the importance of different time steps in the sequence, effectively focusing on the most active regions of spectral change. The architecture concludes with a dense output layer that generates the final prediction. This hierarchical amalgamation of components—justified quantitatively in our ablation study (Section V-B)—enables the model to effectively learn the complex spatio-temporal physics of the SRS process.

E. Model Training and Inference Algorithm

The training and inference procedures define how the model learns from data and subsequently makes predictions. We employed the Adam optimization algorithm to achieve efficient gradient descent. The learning rate was set to $1e^{-5}$, and the model was trained for 100 epochs with a batch size of 64.

During the training phase, the model is trained on labeled data sequences. A sequence of 15 spectral steps ($L = 15$) is input to the model to predict the spectrum of the subsequent time step. The training process is supervised, aiming to minimize the mean squared error (mse) loss function, $\mathcal{L}_\theta$, between the model's prediction and the ground truth from the simulation. The loss function is defined as

$$\mathcal{L}_\theta = \frac{\sum_{i,m,z} (\psi_{i,m,z} - \hat{\psi}_{i,m,z})^2}{\sum_{i,m,z} (\psi_{i,m,z})^2} \quad (4)$$

where $\psi_{i,m,z}$ and $\hat{\psi}_{i,m,z}$ are the true and predicted pulse envelopes, respectively. The index i represents the i th sample, while m and z are the indices for the energy distribution (spectral bins) and propagation steps, respectively. The model parameters θ are updated iteratively via backpropagation to find the optimal set θ_* that minimizes this loss.

Algorithm 1 Training and Inference of the Hybrid CNN-BiLSTM Model

Phase 1: Training

- 1: **Hyperparameters**: Learning rate η , epochs E , batch size B , sequence length $L = 15$.
- 2: **Input**: Training dataset $\mathcal{D} = \{(X_i, y_i)\}_{i=1}^N$.
- 3: Initialize Hybrid Model $\mathcal{F}_\theta$ with parameters θ .
- 4: **for** epoch = 1 to E **do**
- 5: **for** each batch (X_b, y_b) in $\mathcal{D}$ **do**
- 6: $\hat{y}_b \leftarrow \mathcal{F}_\theta(X_b)$ ▷Forward pass
- 7: $\text{loss} \leftarrow \text{MSE}(\hat{y}_b, y_b)$ ▷Compute loss
- 8: Compute gradients $\nabla_\theta \text{loss}$ ▷Backpropagation
- 9: $\theta \leftarrow \text{AdamUpdate}(\theta, \nabla_\theta \text{loss}, \eta)$ ▷Update weights
- 10: **end for**
- 11: **end for**
- 12: **Output**: Trained model parameters θ^* .

Phase 2: Inference (Auto-regressive Prediction)

- 13: **Input**: Initial seed sequence S_{seed} , trained model $\mathcal{F}_{\theta^*}$, prediction steps N_{steps} .
- 14: Initialize full evolution $S_{\text{full}} \leftarrow S_{\text{seed}}$.
- 15: Initialize current sequence $S_{\text{current}} \leftarrow S_{\text{seed}}$.
- 16: **for** $i = 1$ to N_{steps} **do**
- 17: $\hat{y}_{\text{next}} \leftarrow \mathcal{F}_{\theta^*}(S_{\text{current}})$ ▷Predict next spectrum
- 18: Append $\hat{y}_{\text{next}}$ to S_{full} .
- 19: $S_{\text{current}} \leftarrow \text{SlideWindow}(S_{\text{current}}, \hat{y}_{\text{next}})$ ▷Update sequence for next step
- 20: **end for**
- 21: **Output**: Full predicted evolution sequence S_{full} .

To provide a clear, formal, and reproducible description, the step-by-step procedures for both training and inference are detailed in Algorithm 1. In the training phase, the model parameters θ are optimized iteratively. For each training epoch, the dataset is processed in batches. A forward pass is performed using the hybrid model $\mathcal{F}_\theta$, whose internal architecture involving CNN, BiLSTM, and Attention is described in Section III-D and illustrated in Fig. 3. The mse loss between the prediction $\hat{y}_b$ and the ground truth y_b is calculated. Gradients are then computed via backpropagation, and the model weights are updated using the Adam optimizer. In the inference phase, the trained model $\mathcal{F}_{\theta^*}$ predicts the spectral evolution autoregressively. Starting with an initial known sequence S_{seed} , the model predicts the next spectrum $\hat{y}_{\text{next}}$. This prediction is stored, and the input sequence for the next step, S_{current} , is updated by sliding the window forward—removing the oldest spectrum and adding the new prediction. This iterative process continues for the desired number of steps (N_{steps}) to generate the full evolution sequence S_{full} .

F. Complexity Analysis

To understand the theoretical basis for the computational speedup, we analyze the time and space complexity of our proposed model during inference and compare it to that of the traditional SSFM baseline.

1) *Hybrid Model Complexity*: The complexity of our hybrid neural network is determined by its fixed architecture. Let P be the total number of trainable parameters in the model.

1) *Time Complexity*: For a single forward pass (predicting one step), the time complexity is a function of the model's architecture but is constant with respect to the fiber length. To predict an entire evolution of N_{steps} , the model is called auto-regressively N_{steps} times. Therefore, the total time complexity is linear with the number of prediction steps: $O(N_{steps})$.

2) *Space Complexity*: During inference, the space complexity is dominated by the memory required to store the model's parameters. The space complexity is therefore $O(P)$, which is constant with respect to the prediction length.

2) *SSFM Complexity*: The SSFM is an iterative numerical method whose complexity depends on both the fiber length and the required spectral resolution. Let N_z be the number of propagation steps along the fiber and N_t be the number of points for the fast Fourier transform (FFT).

1) *Time Complexity*: Each step involves at least one forward and one inverse FFT, each with a complexity of $O(N_t \log N_t)$. The total time complexity for the SSFM is therefore approximately $O(N_z \cdot N_t \log N_t)$.

2) *Space Complexity*: The SSFM requires storing the complex electric field array at each step, a space complexity of $O(N_t)$.

The fundamental difference in time complexity explains the significant computational advantage of our deep learning approach. To ensure numerical accuracy, the number of SSFM steps, N_z , must be very large (often thousands), whereas our model's required prediction steps, N_{steps} , can be much smaller (e.g., around 100). Furthermore, the SSFM's complexity scales super-linearly with the spectral resolution ($N_t \log N_t$). This theoretical difference in scaling is the primary reason for the substantial speedup (116× on GPU) observed in our results.

IV. DATA ACQUISITION

A. Simulation Data Generation Through SSFM

Due to the prohibitively high costs associated with acquiring a sufficient volume of data for training deep learning models through experimental means, alternative methods such as simulation or synthetic data generation become imperative. Specifically, the CNLSE serves as a fundamental model for describing the evolution of optical pulse propagation in fiber optics. Employing SSFM to numerically solve the CNLSE allows us to simulate the SRS effect of single pulses in a nonlinear optical fiber. The SSFM algorithm is particularly well-suited for this task, as it discretizes the CNLSE into smaller steps, making it computationally efficient and allowing for the simulation of pulse propagation under various conditions.

The CNLSE can be described by [19]

$$\frac{\partial A_p}{\partial z} + \frac{i\beta_{2p}}{2} \frac{\partial^2 A_p}{\partial T^2} = i\gamma_p \left[|A_p|^2 + (2 - f_R) |A_s|^2 \right] A_p - \frac{g_p}{2} |A_s|^2 A_p \quad (5)$$

$$\frac{\partial A_s}{\partial z} - d \frac{\partial A_s}{\partial T} + \frac{i\beta_{2s}}{2} \frac{\partial^2 A_s}{\partial T^2} = i\gamma_s \left[|A_s|^2 + (2 - f_R) |A_p|^2 \right] A_s + \frac{g_s}{2} |A_p|^2 A_s \quad (6)$$

where A_p and A_s denote the slow-varying envelopes associated with the pump and Raman pulses, respectively; β_{2p} and β_{2s} represent the group velocity dispersion (GVD) coefficients for the pump and Stokes pulses, respectively; γ_p and γ_s are their corresponding nonlinear coefficients; and g_p and g_s correspond to the Raman gain coefficients at the pump and Stokes wavelengths. The parameter f_R represents the fractional Raman contribution. In (5) and (6), $T = t - z/v_{gp}$ defines the retarded frame, which serves as a reference frame moving with the pump pulse at the group velocity (v_{gp}). The parameter $d = v_{gp}^{-1} - v_{gs}^{-1}$ accounts for the walk-off effect, representing the group velocity mismatch between the pump and Stokes pulses. Here, v_{gs} represents the group velocity of the Stokes wave. It is worth noting that the fiber loss has been neglected in (5) and (6) due to the relatively short length of the fiber used in our experiments. The fractional Raman contribution f_R of the As_2S_3 glass is 0.031, and the Raman gain spectrum is characterized by a primary Stokes shift of approximately 10.3 THz [39].

The wavelength-dependent GVD coefficients (β_{2p} and β_{2s}) and nonlinear coefficients (γ_p and γ_s) used in the simulation were derived from the calculated curves presented in Fig. 4(a) and (b), respectively. The fabrication process of the As_2S_3 MOF utilized in these experiments is detailed in Supplementary Material (Section I). Fig. 4 illustrates the key parameters and information derived from the laboratory fabrication of the As_2S_3 MOF. The cross section of the optical fiber was measured with an optical microscope, revealing a core diameter of approximately $3.23 \mu\text{m}$, defined as the diameter of the inscribed circle within the triangular core. Hole sizes range from 30 to $50 \mu\text{m}$, and the outer cladding diameter is around $160 \mu\text{m}$. Using the finite difference method, we modeled the actual cross section of the optical fiber. The glass refractive index was calculated using Sellmeier coefficients from reference [39], determining the effective refractive index (n_{eff}), GVD parameter β_2 , nonlinear coefficient, and effective mode area (A_{eff}), as depicted in the respective figures. The inset in Fig. 4(b) illustrates the mode field distribution of the fundamental mode at 2500 nm, with an effective mode area of approximately $8.44 \mu\text{m}^2$. The ZDW is approximately $2.46 \mu\text{m}$. This simulation setup defined our core dataset parameters: a temporal window of 18 ps represented by 1024 discrete points, which directly corresponds to the 1024 input and output features of our hybrid model.

B. Experiments of SRS in As_2S_3 MOF

To further validate the accuracy of our model's predictions, we conducted a series of experiments and collected corresponding data for comparison. Fig. 5 illustrates the experimental arrangement employed to induce SRS effects in suspended-core As_2S_3 MOF. A picosecond pulse source served as the pump, emitting pulses with an FWHM of 2.7 ps and a repetition rate of 31.94 MHz at a wavelength of 1958 nm. The pump pulses were directed into As_2S_3 MOF through

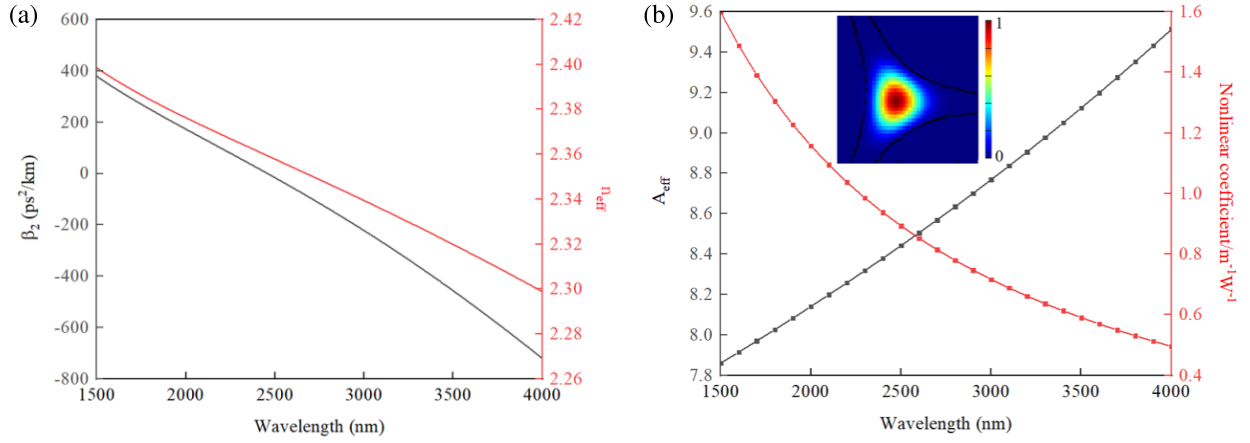


Fig. 4. (a) Calculated β_2 and n_{eff} of the fundamental mode from 1500 to 4000 nm. (b) Calculated A_{eff} nonlinear coefficient of the fundamental mode from 1500 to 4000 nm. The inset shows the fundamental mode field distribution at 2500 nm. For detailed preparation procedures, please refer to the Supplementary Material.

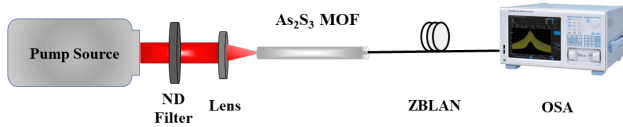


Fig. 5. Experimental setup for SRS effects in suspended-core As₂S₃ MOF. Refer to the Supplementary Material for details on the preparation of As₂S₃ MOF.

a neutral density (ND) filter and a focusing lens (Thorlabs C028TME-D) with a focal length of 5.95 mm and a numerical aperture of 0.56 at the design wavelength of 4.1 μm . The transmission of the focusing lens at 1958 nm was measured to be 97.7%. To ensure optimal cross sections, the ends of As₂S₃ MOF were meticulously cleaved using a diamond stylus. The coupling efficiency into the As₂S₃ MOF was carefully measured and incorporated into our power scaling analysis to ensure a valid comparison between experimental results and numerical simulations. The efficiency was determined by measuring the average power before the input objective lens and the transmitted power at the fiber output. After accounting for the calculated Fresnel reflection losses at the As₂S₃-to-air interfaces (refractive index $n \approx 2.4$, yielding $R = ((n - 1)/(n + 1))^2 \approx 17\%$ per facet), the actual power exiting the fiber core ($P_{\text{out_core}} = P_{\text{out_raw}}/(1 - R)$) and the power entering the fiber core ($P_{\text{in_core}} = P_{\text{in_before_lens}} \times (1 - R)$) were estimated. The overall coupling efficiency of the pump light into the fiber core was then calculated as $\eta = P_{\text{out_core}}/P_{\text{in_core}}$, determined to be approximately 25%. This measured efficiency was crucial for mapping the experimental conditions to the simulation parameters. The pump powers reported in our experiments correspond to the average power (P_{av}) measured at the laser source output. To obtain the peak power (P_{pk}) inside the fiber—the parameter required for the CNLSE simulations—we used the following two-step procedure. First, the coupled average power was calculated as $P_{\text{av,coupled}} = P_{\text{av}} \times \eta$, where $\eta = 0.25$ is the coupling efficiency. Second, this value was converted to peak power using the relation

$$P_{\text{pk}} = \frac{P_{\text{av,coupled}}}{f \times t} \quad (7)$$

where f is the laser's repetition rate (31.94 MHz) and t is the pulse FWHM (2.7 ps). For example, a measured average laser power of 81 mW corresponds to a peak power of about 235 W for the simulation. This calculation was applied to all experimental power levels. The output ends of As₂S₃ MOF were directly linked to a 0.5-m large-mode-area (LMA) ZBLAN fiber with a core diameter of 105 μm and a transmission window spanning from 0.4 to 5.0 μm . The considerable core diameter of the LMA ZBLAN fiber mitigated nonlinear effects. Spectral data were captured by an optical spectrum analyzer (OSA) (Yokogawa AQ6375) connected to the LMA ZBLAN fiber. A 1958-nm pump signal pulse was introduced into a 10-cm-long As₂S₃ MOF at various average pump powers, and the resulting output SRS spectra are shown in Fig. 6(a). Below 81 mW, no observable SRS output spectrum was detected, indicating that the pump power was below the threshold. At an average pump power of 81 mW, the first-order forward Stokes (R_{S1}) appeared at 2100 nm, while the first-order anti-Stokes (R_{AS1}) appeared at 1840 nm, marking the power threshold for SRS in the suspended-core As₂S₃ MOF. As the average pump power increased to 116 mW, the intensities of R_{S1} and R_{AS1} continued to rise. This trend persisted as the average pump power was further increased from 116 to 240 mW. Concurrently, due to the combined effects of the inherent Raman gain spectrum and self-phase modulation (SPM) of the MOF, the widths of R_{S1} and R_{AS1} broadened. At 240-mW average pump power, R_{S1} and R_{AS1} broadened to form a supercontinuum spectrum. Similarly, when the signal pulse was introduced into the 30-cm-long As₂S₃ MOF, the spectral trend was similar to that of the 10-cm As₂S₃ MOF, as shown in Fig. 6(b). However, due to the low experimental power, the higher order SRS generated were not as prominent. Nevertheless, Fig. 6 demonstrates that the SRS gain spectrum ranges from 1400 to 2400 nm, providing an opportunity to explore stable Raman amplifiers over a wide wavelength range.

V. RESULTS

The model's performance is quantitatively measured throughout this section using a root-mean-squared error

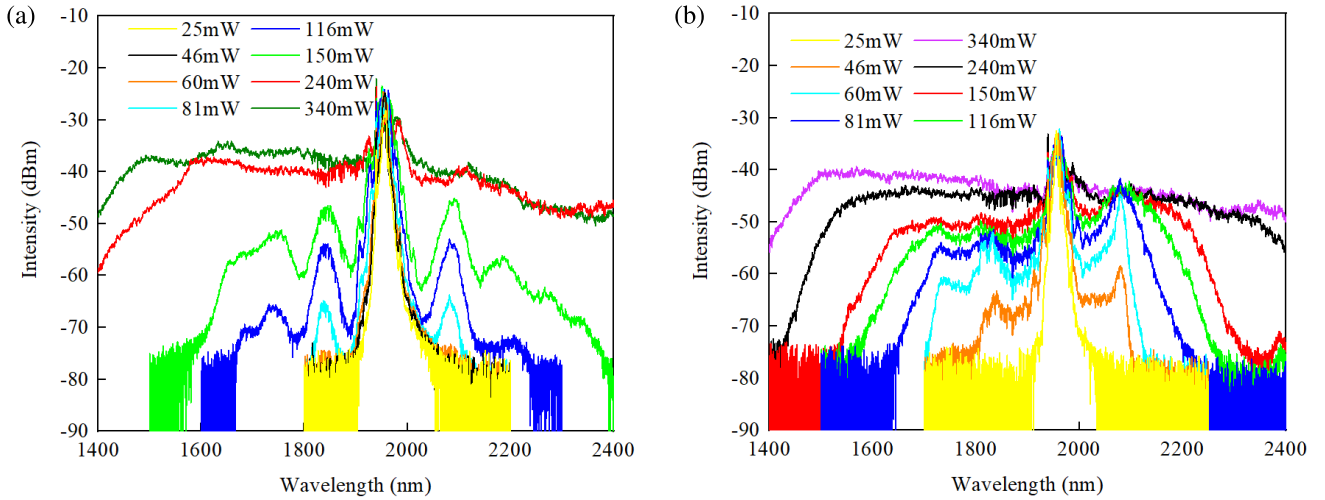


Fig. 6. Measured Raman spectra in the As_2S_3 MOF with different pump powers in (a) 10-cm-long fiber and (b) 30-cm-long fiber.

(RMSE) metric, $\mathcal{R}$, defined as

$$\mathcal{R} = \sqrt{\frac{\sum_{m,z} (X_{SSFM} - X_{pred})^2}{\sum_{m,z} (X_{SSFM})^2}} \quad (8)$$

where X_{SSFM} represents the ground truth intensity distributions from the CNLSE simulation and X_{pred} represents the corresponding intensity distributions predicted by our hybrid model.

A. Model Performance on As_2S_3 MOF

We initially trained the hybrid model to simulate the propagation of picosecond pulses within the normal dispersion range of highly nonlinear chalcogenide fibers. The dataset comprises 1000 NLSE numerical simulations of propagation over 10 cm of fiber, employing initial conditions of input Gaussian pulses with peak power P_0 uniformly distributed over the range of 200–500 W. Therefore, considering the coupling efficiency of the fiber in experiments, our actual range of pump power is 69–172 mW. The 1000 simulations were allocated into sets of 800 for training, 100 for validation, and 100 for testing. In the simulation data, to align more closely with experimental measurements, we also accounted for the influence of other nonlinear effects such as SPM and self-steepening. Fig. 7 depicts the results obtained during the training of the network to model the spectral intensity evolution, contrasting with the evolution of the spectral intensity simulated using the CNLSE. The specific results showcased correspond to peak powers of 81, 116, and 150 mW, stimulating first- to third-order Stokes and anti-Stokes. The overall visual agreement between the propagation dynamics predicted by the hybrid model and those simulated from the CNLSE is remarkable. For ease of visualization, the evolution is plotted on a logarithmic scale. Additionally, it is noteworthy that the Raman frequency shifts and associated spectral intensity profiles are particularly well-predicted by the model. The right of Fig. 7 displays the relative difference between the CNLSE and model evolution maps. Comparisons between CNLSE and model evolution for 100

different input conditions spanning the full range of parameter variation yielded similar results, with an RMSE computed over the 100 evolution maps of $\mathcal{R} = 0.0558$. A more detailed comparison between CNLSE simulations, model predictions, and experimental measurements at the fiber output (10 cm) is provided in Fig. 8. These results directly validate the model's ability to learn the core physics of the SRS process. Based on the 10.3-THz (344 cm^{-1}) Raman shift of As_2S_3 and the 1958-nm pump, the R_{S1} and R_{AS1} peaks are expected at approximately 2100 and 1840 nm, respectively. As shown in Fig. 8(a) at 81 mW, the CNLSE simulation, the hybrid model prediction (red dashed line), and the experimental measurement (black dotted line) all precisely align at these 2100 nm (R_{S1}) and 1840 nm (R_{AS1}) locations. As the power increases [116 mW, Fig. 8(b), and 150 mW, Fig. 8(c)], the hybrid model continues to accurately predict the exact spectral locations of all experimentally observed higher order components, demonstrating its high precision in learning the fundamental Raman shifts. However, while frequency prediction remains accurate, quantitative discrepancies arise in the predicted intensities, particularly for higher order peaks at high power. In the 150-mW case [Fig. 8(c)], the predicted peak intensity of the R_{S1} shows good agreement, with a deviation of less than 3 dB from the experiment. In contrast, the predicted intensity of the R_{S2} is underestimated by approximately 15 dB compared to the experimental measurement. This suggests that while our model accurately learns the underlying physics governing the spectral locations of SRS, the complex energy distribution dynamics at high intensities involve factors potentially beyond the scope of the standard CNLSE simulation used for training. A detailed discussion of the physical origins of this intensity discrepancy is provided in Section VI-A. To evaluate the robustness of the hybrid model, we extended the fiber length to 1 m and examined the effects of varying both the input peak power and pulse duration of the laser simultaneously. Considering the coupling efficiency of the optical fiber, the actual input average power ranged from 3 to 65 mW, and the FWHM pulsewidth ranged from 1 to 3 ps. We allocated 800 datasets for training, 100 datasets for validation, and the remaining 100 datasets

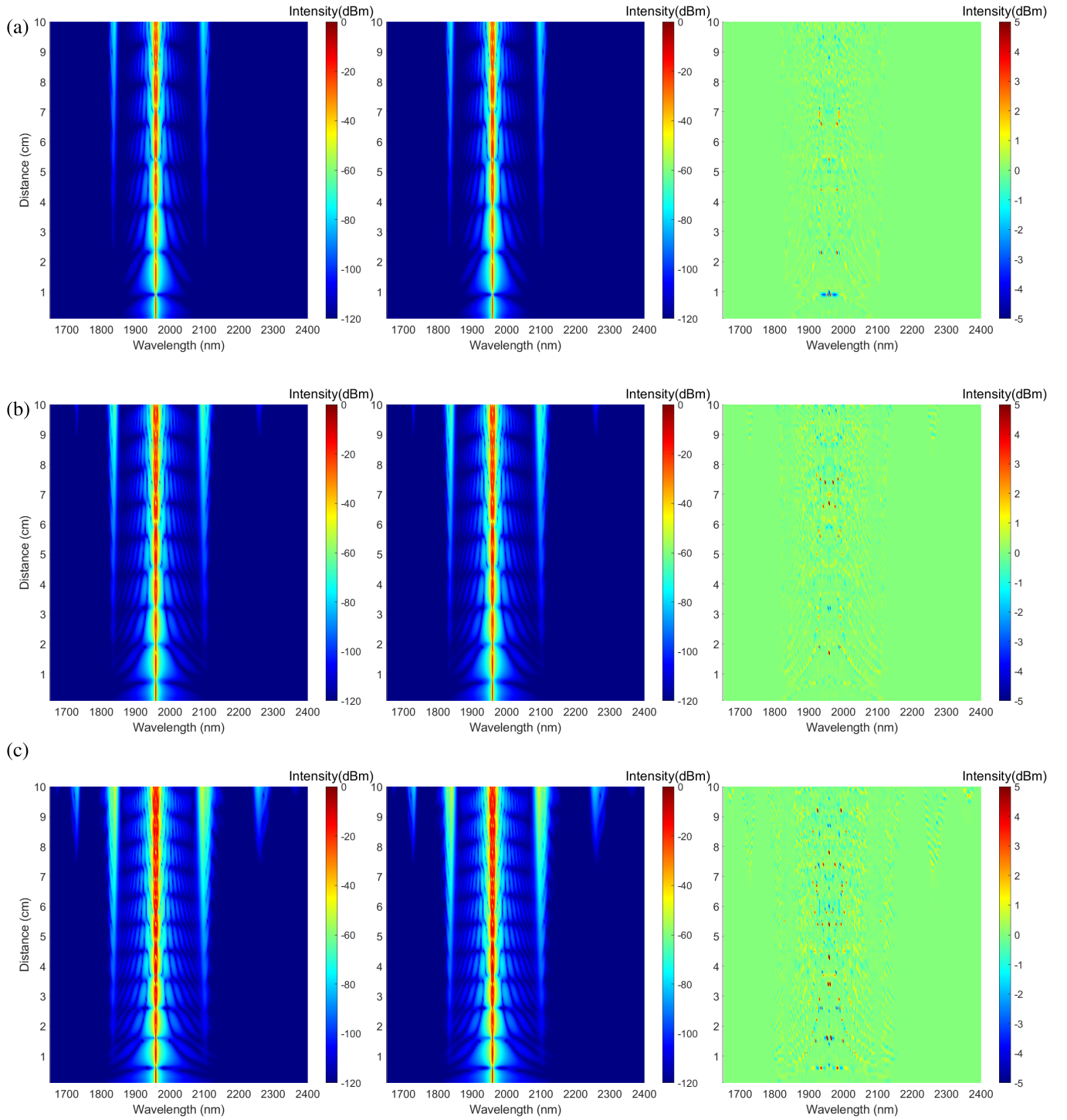


Fig. 7. Hybrid modeling of spectral evolution in a 10-cm As_2S_3 MOF under different average pump powers. (a) Pumping at 81 mW. (b) Pumping at 116 mW. (c) Pumping at 150 mW. Left: numerical simulations (CNLSE). Middle: hybrid neural network predictions. Right: error maps between simulations and predictions. In the left and middle, the gradient from blue to red indicates increasing light intensity. In the right, green (0 dBm) indicates a close match between the predicted and simulated values.

for testing. Fig. 9 illustrates the evolution maps of the trained hybrid model and CNLSE simulations across different laser input powers and FWHM pulsewidths, along with detailed spectral diagrams of the CNLSE and hybrid model at 1 m. Again, there is an excellent visual agreement between the CNLSE simulations and the hybrid model predictions. The

hybrid model effectively captures the Raman gain spectra at different powers and pulsewidths and accurately predicts SRS effects from the first order to higher orders, demonstrating its strong generalization ability. A comparison of CNLSE and model evolution maps for 100 different input conditions (covering the entire range of parameter variations and not used

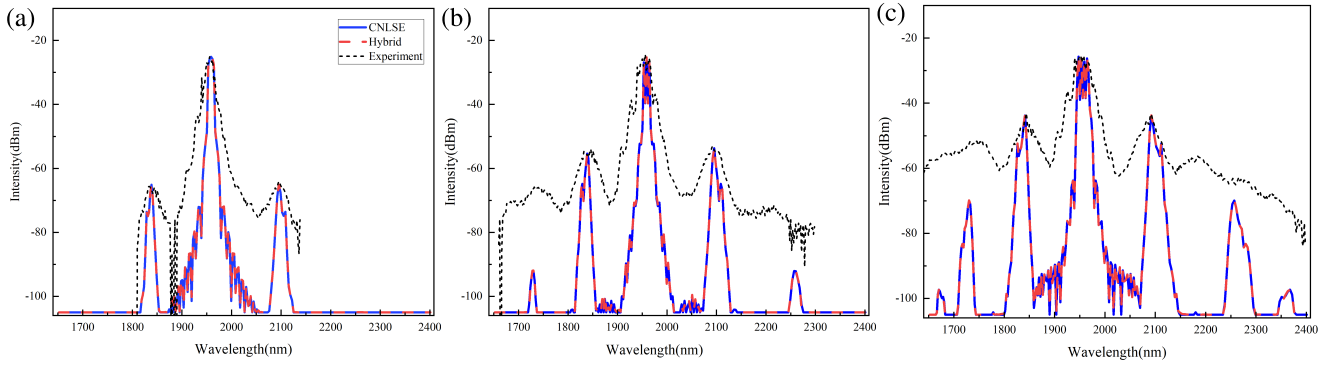


Fig. 8. Comparison of Raman spectra at 10-cm As_2S_3 MOF simulated with the CNLSE (blue solid lines), predicted by the hybrid model (red dashed lines), and experimentally measured (black dotted lines) under different average pump powers. (a) By 81-mW pumping. (b) By 116-mW pumping. (c) By 150-mW pumping.

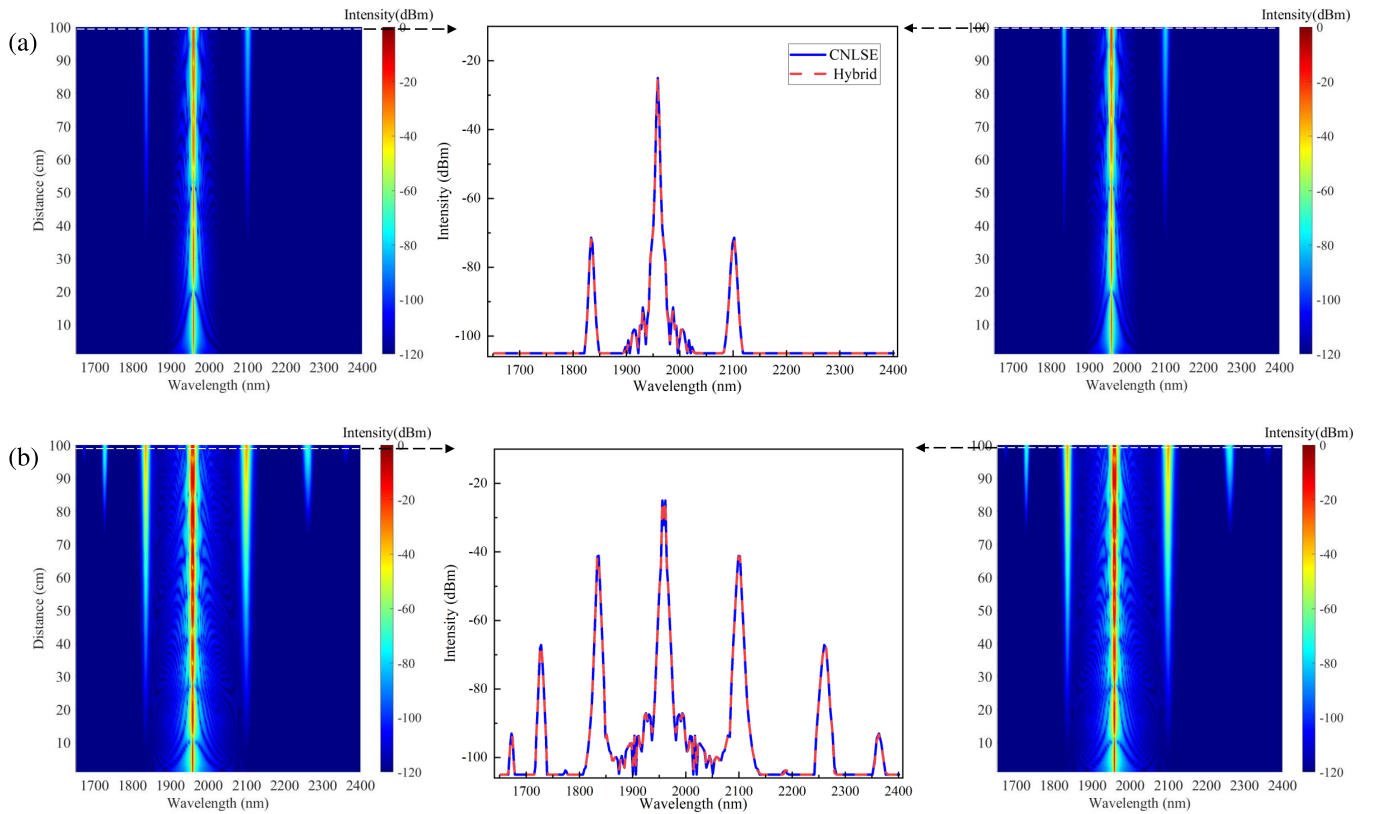


Fig. 9. Hybrid modeling of spectral evolution in a 1-m As_2S_3 MOF under different average pump powers and FWHMs. (a) Pumping at 20 mW and FWHM is 1 ps. (b) Pumping at 55 mW and FWHM is 1.5 ps. Left: numerical simulations (CNLSE). Right: hybrid neural network predictions. Middle: comparison between simulated (blue solid lines) and predicted (red dashed lines) profiles at the 1-m output of the As_2S_3 optical fiber. In the left and right, the gradient from blue to red indicates increasing light intensity.

in the training phase) yielded similar results, with an RMSE of $\mathcal{R} = 0.0826$ calculated over the 100 evolution maps.

B. Model Performance on As_2Se_3 MOF

To further validate the applicability and robustness of our proposed hybrid neural network architecture, we conducted an additional case study on a physically distinct material: As_2Se_3 MOF. As a prominent member of the chalcogenide glass family, As_2Se_3 possesses significantly different physical properties from As_2S_3 , including a much higher nonlinear refractive index (approximately 3–4 times greater) and a

distinct Raman gain spectrum. The primary Raman shift in As_2Se_3 is centered at approximately 7.2 THz ($\sim 240 \text{ cm}^{-1}$) [40], in contrast to the $\sim 10.3 \text{ THz}$ ($\sim 344 \text{ cm}^{-1}$) shift in As_2S_3 . A new dataset was generated using the SSFM procedure (as described in Section IV-A). The fractional Raman contribution f_R of the As_2Se_3 glass is 0.148 [40]. This dataset was designed to test the model under diverse conditions. Specifically, we generated 1000 sets for a 1-m As_2Se_3 MOF, with input average powers ranging from 0- to 10-mW and FWHM pulsewidths from 1 to 3 ps. We also generated 1000 sets for a 10-cm As_2Se_3 MOF, with input average powers ranging from 10 to

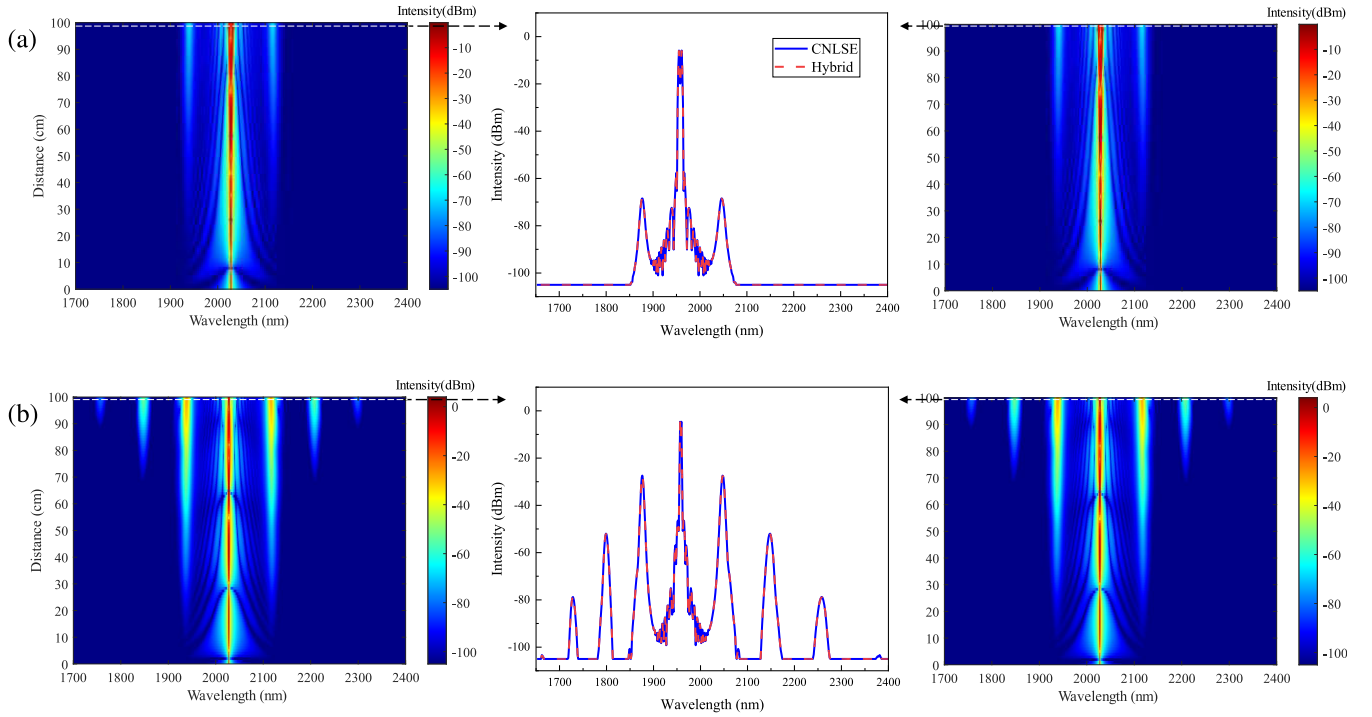


Fig. 10. Hybrid modeling of spectral evolution in a 1-m As_2Se_3 MOF under different average pump powers and FWHMs. (a) Pumping at 2 mW and FWHM is 1 ps. (b) Pumping at 8 mW and FWHM is 2.5 ps. Left: numerical simulations (CNLSE). Right: hybrid neural network predictions. Middle: comparison between simulated (blue solid lines) and predicted (red dashed lines) profiles at the 1-m output of the As_2Se_3 optical fiber. In the left and right, the gradient from blue to red indicates increasing light intensity.

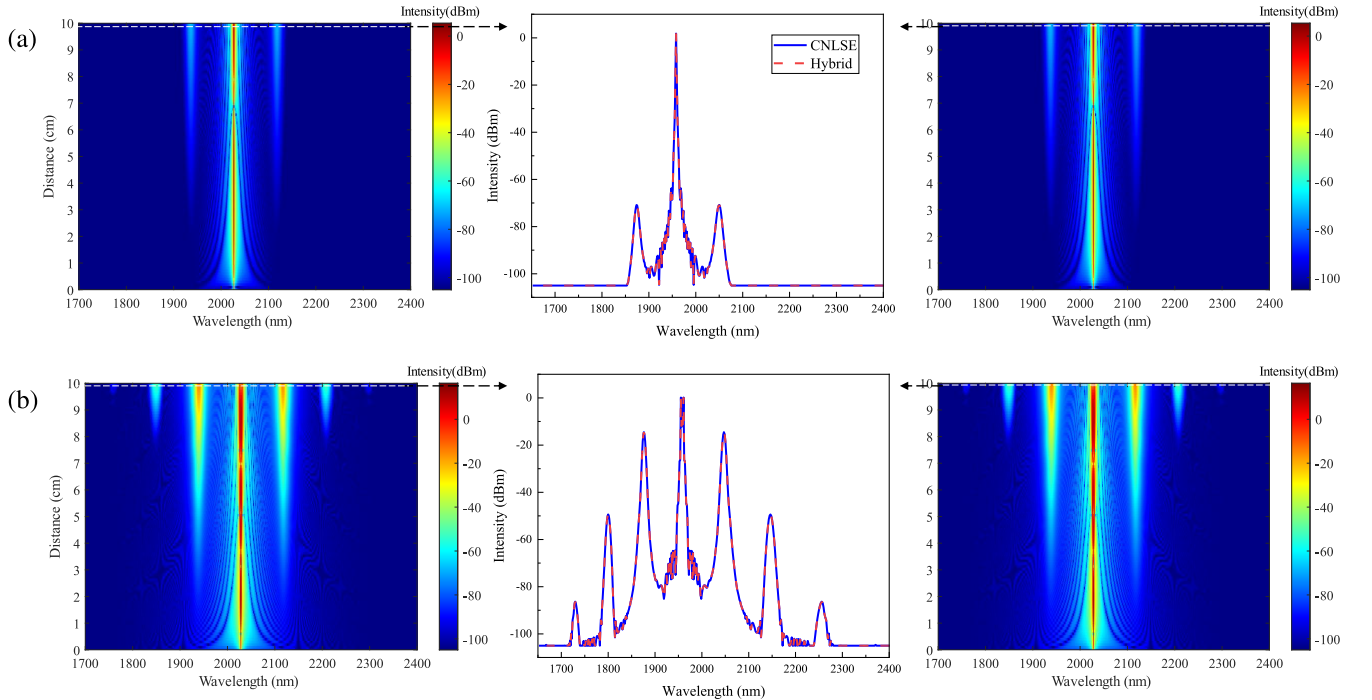


Fig. 11. Hybrid modeling of spectral evolution in a 10-cm As_2Se_3 MOF under different average pump powers and FWHMs. (a) Pumping at 11 mW and FWHM is 1 ps. (b) Pumping at 38 mW and FWHM is 2.5 ps. Left: numerical simulations (CNLSE). Right: hybrid neural network predictions. Middle: comparison between simulated (blue solid lines) and predicted (red dashed lines) profiles at the 10-cm output of the As_2Se_3 optical fiber. In the left and right, the gradient from blue to red indicates increasing light intensity.

50 mW and FWHM pulsewidths from 1 to 3 ps. For validation (10%), and testing (10%) sets, identical to the both datasets, the data were partitioned into training (80%), methodology used for the As_2S_3 data as in Section V-A.

TABLE I
ABLATION STUDY RESULTS ON THE TEST SET

Model Configuration	RMSE	Relative Performance Drop (%)
Full Model (2D CNN + BiLSTM + Attention)	0.0859	–
w/o Attention (2D CNN + BiLSTM)	0.0992	15.6%
1D CNN + BiLSTM + Attention	0.1138	32.6%
w/o CNN (BiLSTM + Attention)	0.1289	50.1%
BiLSTM only	0.1410	70.7%
w/o BiLSTM (2D CNN + Attention)	0.2095	144.0%
2D CNN only	0.2619	205.1%

The hybrid model was then trained from scratch on this new As_2Se_3 dataset using the identical architecture and training procedure described in the main manuscript. Fig. 10 (1-m MOF) and Fig. 11 (10-cm MOF) illustrate the evolution maps of the trained hybrid model and CNLSE simulations across different laser input powers and FWHM pulsewidths, along with detailed spectral comparisons at the fiber end. Again, there is an excellent visual agreement between the CNLSE simulations and the hybrid model predictions. The hybrid model effectively captures the distinct Raman gain spectra of As_2Se_3 , accurately predicting the new spectral locations (e.g., R_{S1} at ~ 2055 nm) and the evolution of SRS effects from the first order to higher orders. This demonstrates the strong applicability of our architecture to a physically different system. To quantitatively validate this, a comparison of CNLSE and model evolution maps for the 100 unseen test conditions in each dataset was performed. For the 1-m As_2Se_3 dataset, this yielded an rms error of $\mathcal{R} = 0.0891$, while the 10-cm dataset yielded $\mathcal{R} = 0.0615$. This successful validation confirms that our proposed framework is a robust tool for modeling complex SRS dynamics across different chalcogenide materials.

C. Ablation Study

To validate the effectiveness of our proposed hybrid architecture and demonstrate the contribution of each key component, we conducted a comprehensive ablation study. We compared the performance of our full model (2-D CNN + BiLSTM + Attention) against several ablated versions and baselines, as shown in Table I. To ensure a sufficiently complex evaluation scenario, all model variants were trained and tested under identical conditions on the 1-m As_2S_3 and As_2Se_3 datasets, which feature variations in both pump power and pulse duration. The performance was evaluated using the RMSE on the unseen test set, and the results are summarized in Table I.

The results in Table I clearly demonstrate that a synergistic hybrid architecture is essential for this task. The full model achieves the lowest RMSE of 0.0859, while models using only a single component perform very poorly; the 2-D CNN-only model (205.1% performance drop) and the BiLSTM-only model (70.7% drop) are insufficient on their own. This confirms that the problem is fundamentally spatio-temporal, requiring both a spatial feature extractor and a temporal sequence processor.

The analysis reveals a clear hierarchy of component importance. The BiLSTM, which models the temporal evolution,

is the most critical component; removing it (w/o BiLSTM) causes a catastrophic 144.0% performance drop. However, the BiLSTM requires a feature extractor to function effectively. Removing the CNN (w/o CNN) causes the BiLSTM to struggle with the high-dimensional raw data, resulting in a significant 50.1% error increase.

Most importantly, this study quantitatively validates our novel 2-D CNN approach. The standard 1-D CNN baseline results in a 32.6% performance drop compared to our full model. This proves that our strategy of reshaping the input data into a 2-D tensor allows the 2-D CNN to capture complex spatio-temporal features that a standard 1-D CNN misses. Finally, removing the attention mechanism (w/o Attention) causes a 15.6% drop, confirming that it provides a valuable refinement to the model's predictions. This ablation study provides strong quantitative evidence that our proposed hybrid architecture is a well-designed and effective solution for the task.

D. Computational Performance

The accuracy of traditional numerical solution methods is heavily reliant on the number of grid points, and as the number of grid points increases, so does the calculation and solution time. For instance, using the SSFM-based CNLSE solution, it takes 393.26 s to generate 100 sets of simulation data. However, the proposed model offers a solution to this challenge. Once trained, this model only requires 3.399 s to generate the same amount of simulation data on a GPU, marking a speed increase of approximately 116 times. Even when utilizing a CPU for simulation, the time reduces to 8.908 s, which represents over a 44-fold improvement in simulation speed. Moreover, as the numerical solution employs more grid points for simulation, the efficiency of the hybrid model can be further enhanced. All codes presented in this article were developed using Python 3.10, employing TensorFlow 2.10, and MATLAB R2022a. The numerical examples were executed on a system equipped with a 3.4-GHz 16-core Intel Core i7 processor and an NVIDIA GeForce RTX 4080 graphics processing unit.

VI. DISCUSSION

A. Interpretation of Model Performance

The presented results strongly validate our hybrid deep learning architecture as a powerful surrogate for modeling the complex dynamics of SRS in chalcogenide MOFs. The

model demonstrates high fidelity to both SSFM simulations (e.g., $\mathcal{R} = 0.0558$ for the 10-cm fiber) and experimental measurements, particularly for primary spectral features. The most significant practical implication is the transformation of SRS modeling from a computationally intensive, offline task into a near-real-time capability, achieved through a remarkable speedup of $116\times$ on a GPU compared to SSFM. This acceleration opens new avenues for rapid parameter space exploration and experimental optimization.

Despite the model's overall success, particularly in predicting the exact spectral locations of all observed SRS peaks across all power levels (Fig. 8)—confirming its grasp of the fundamental Raman shifts—a systematic discrepancy in predicted intensities emerges at high pump powers. As quantified in Section V-A, while the deviation for the first-order Stokes peak remains modest (<3 dB at 150 mW), the intensities of higher order peaks (e.g., R_{AS2}) are significantly underestimated by 15 dB compared to the experimental data.

We argue that this intensity mismatch likely originates not primarily from a failure of the neural network to learn the training data, but rather from the inherent complexity of nonlinear optical interactions at high intensities within the MOF, which may not be fully captured by the standard CNLSE model used to generate the SSFM training data. Several physical factors likely contribute to this deviation. First, while the CNLSE includes SRS, the precise efficiency of cascading SRS (where Stokes waves pump higher orders) might be sensitive to parameters or subtle effects not perfectly modeled in the simulation. Second, and potentially more significantly, at high peak powers and broad spectral bandwidths, coexisting nonlinear effects, particularly four-wave mixing among the pump, Stokes, and anti-Stokes waves, become increasingly important [4], [6]. The intricate interplay and competition between SRS and FWM, strongly influenced by the MOF's specific dispersion profile, can redistribute energy among spectral components in complex ways that a standard CNLSE simulation might only approximate. Capturing these combined effects perfectly represents a frontier challenge in nonlinear optics modeling. Therefore, the observed intensity discrepancy underscores this physical complexity, while the model's success in predicting spectral frequencies highlights its robust learning of the core SRS physics. This points toward future work involving training models on data from more sophisticated simulations or directly on experimental data to potentially capture these richer dynamics.

B. Significance and Contribution

The primary contribution of this work is the novel application and validation of a synergistic hybrid deep learning architecture for a new and challenging problem in nonlinear optics. While prior studies have successfully applied deep learning to phenomena like supercontinuum generation [20], [21], this study, to our knowledge, presents the first framework specifically designed and experimentally validated for MIR SRS dynamics in chalcogenide MOFs. Our ablation study quantitatively confirmed that the hierarchical design—using a CNN for spatial feature extraction followed by a BiLSTM

for temporal sequence modeling—is essential for accurately learning the complex physics of SRS evolution.

This high-efficiency framework has significant implications for the photonics community. It enables the rapid exploration of vast parameter spaces, a critical step toward realizing inverse design, where models can be used to engineer fibers and pulse conditions to achieve specific spectral outputs. Crucially, by validating our model against the first experimental demonstration of 2- μm pumped SRS in a suspended-core As_2S_3 MOF, our work bridges the critical gap between numerical modeling and real-world application, enhancing the practical relevance and trustworthiness of deep learning approaches in experimental photonics. Moreover, the robustness and generalizability of our architecture were rigorously confirmed through an additional case study on a physically distinct As_2Se_3 MOF. This successful validation against a different simulated material system proves our framework's capability to adapt to varied physical parameters, solidifying its role as a powerful and adaptable design tool.

C. Limitations and Future Work

We acknowledge several limitations in the current study, which open avenues for future research. First, the model was trained primarily on simulation data, with experimental data used only for validation. As discussed, this means that the model's accuracy is fundamentally bounded by the accuracy of the SSFM simulation, which led to discrepancies at high pump powers. A promising future direction is to explore transfer learning techniques, where a model pretrained on simulation data could be fine-tuned with a smaller set of experimental data. This approach could potentially allow the model to learn and correct for the simulation–experiment mismatch.

Second, the current model is specific to the geometry and material properties of the chalcogenide MOF used in this study. To create a more generalized tool, a future iteration of the model could be designed to accept fiber parameters (e.g., core diameter, dispersion coefficients, and nonlinear index) as additional inputs. This would enable the network to learn the relationship between fiber properties and the resulting SRS dynamics, creating a powerful and adaptable design tool applicable across a range of different fiber configurations, a concept that has shown promise in related fields. Such advancements would significantly enhance the model's practical utility for the broader photonics community.

VII. CONCLUSION

In summary, this work establishes a validated deep learning framework that fundamentally alters the paradigm of modeling complex nonlinear optical phenomena. We have demonstrated that by synergistically integrating a convolutional front end with a recurrent core and an attention mechanism, it is possible to create a surrogate model for SRS that is not only extraordinarily fast—achieving a 116-fold speedup on a GPU—but also robustly accurate. Crucially, this model's predictive power was confirmed not just against numerical simulations, but also through its validation against the first experimental demonstration of MIR SRS in a suspended-core chalcogenide MOF.

The implications of this validated framework are profound. It transforms the process of experimental design in nonlinear photonics from a slow, iterative cycle of simulation and measurement into a rapid, dynamic loop of in silico exploration and targeted optimization. This acceleration is a critical enabler for the field, paving the way for data-driven inverse design and speeding up the discovery of novel MIR light sources and photonic devices with tailored spectral properties. Our work effectively bridges the gap between theoretical modeling and practical application, enhancing the trustworthiness and utility of AI-driven tools in experimental science.

Looking forward, while this study provides a robust proof of concept, it also illuminates a clear path for future innovation. The next generation of these models will likely evolve toward greater generality by incorporating physical fiber parameters as direct inputs and will leverage transfer learning to fine-tune their predictions with limited experimental data, thereby overcoming the intrinsic limitations of purely simulation-based training. Ultimately, this research demonstrates the potential of hybrid deep learning architectures to serve not merely as computational accelerators but as integral partners in the process of scientific discovery in nonlinear optics.

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Qi Wang received the master's degree in the major of microelectronics and solid-state electronics from Northeastern University, Shenyang, China, in 2023.

Her research interests include optical fiber sensors, nonlinear optics, and soft glasses.



Jun Fu (Senior Member, IEEE) received the Ph.D. degree in mechanical engineering from Concordia University, Montreal, QC, Canada, in 2009.

He was a Post-Doctoral Researcher with the Department of Mechanical Engineering, Massachusetts Institute of Technology (MIT), Cambridge, MA, USA, from 2010 to 2014. In 2015, he became a Professor with the State Key Laboratory of Synthetical Automation for Process Industries, Northeastern University, Shenyang, China, and has been with the State Key Laboratory

since leaving MIT. His current research is on dynamic optimization, machine learning, switched systems, and their applications.

Dr. Fu is a fellow of Chinese Association of Automation. He is an Associate Editor of *Control Engineering Practice*, IEEE TRANSACTIONS ON INDUSTRIAL INFORMATICS, and IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS.



Meiting Pan received the B.A. degree from the College of Physical Science and Technology, Bohai University, Jinzhou, China, in 2021. She is currently pursuing the master's degree with Northeastern University, Shenyang, China.

Her primary research focus is on the nonlinear effects and sensing applications of optical fibers.



Yupeng Liu received the B.A. degree from Northeastern University, Qinhuangdao, China, in 2019, and the M.E. degree from the University of Ottawa, Ottawa, ON, Canada, in 2021. He is currently pursuing the Ph.D. degree in control science and engineering with Northeastern University, Shenyang, China.

His current research focuses on AI for science, deep learning, and photonic crystal fibers.



Tonglei Cheng received the Ph.D. degree in optical engineering from Tianjin University, Tianjin, China, in 2010.

Since 2011, he has been a JSPS Post-Doctoral Fellow with Toyota Technological Institute, Nagoya, Japan. In 2016, he joined Toyota Technological Institute as a Commissioned Scientist. In 2017, he joined Northeastern University, Shenyang, China, as a Full Professor. His current research interests include optical fiber sensors, nonlinear optics, optical fibers, and soft glass fibers.